

# Humans Are More Diverse: Audited Frontier LLMs Show Extreme Policies in Idealised AI Development Races

Anonymous Author(s)  
Submission Id: TBD\*\*\*\*\*†

## ABSTRACT

In an AI development race a company can move slowly and safely, or faster while risking the loss of its final reward. We use this repeated game to study strategic safety behaviour in large language model (LLM) agents. A valid action does not show that an agent understands the game, so an audit gate comes before behavioural reading: it verifies the engine, then tests rule recall, state tracking, payoff calculation and stability under equivalent task wordings. We audit nine frontier routes on one frozen probe bank. Every route reads the payoff matrix perfectly, yet accuracy at rebuilding the current state ranges from 100% to 20%, and five pass. Across their 5,022 gameplay decisions the action format never failed, so a well-formed action says nothing about whether the agent tracked the state. State-reconstruction accuracy is descriptively associated with how sharply play responds to assigned risk, which spans 7.0 to 100.0 percentage points, though passing and failing routes overlap. Presentation matters too: swapping which opaque code denotes SAFE raises UNSAFE play by 8.3 and 9.5 points on the two crossed routes, mechanism unchanged. On raw action sequences, human participants are more diverse than every route that passes the gate, and the only two routes reaching their range both fail it. AI-race simulations therefore need validity checks and trajectory-level analysis before their outputs are called strategic, human-like or safety-aware. Our findings apply only to the tested routes, prompts and hosted decoding contract, which did not forward the requested temperature.

## KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, task validity

## 1 INTRODUCTION

Competition over advanced artificial intelligence can create a safety dilemma: each company may benefit from moving faster, but safety work slows it down when rivals do not follow the same standard [5, 6, 10]. An *AI development race* is a simple game that represents this tension, in which every player chooses each round between slower Safe development and faster Unsafe development [11, 16]. The game is idealised and does not model a real AI company in full. Its value is control: it lets us test how choices change with competition, relative progress, and the earlier actions of other players.

Human experiments show why the order of actions matters. In the two-player study of Domingos and Han [14], the preregistered comparison between the two higher private-risk conditions found no clear difference; exploratory analyses instead linked later Unsafe choices to the opponent’s previous action, to whether the player was ahead or behind, and to the player’s first action. We call this *path dependence*: an early choice or event changes the later path of the interaction. A useful human and LLM comparison must

therefore examine full action sequences, which we call *trajectories*, rather than only the overall Unsafe rate.

Large language models (LLMs) can act as language-based players in this game, and are already studied as simulated participants, decision-making agents and game-playing systems [1, 2, 4]. Their use creates a measurement problem. A model may return a correctly formatted action without tracking the state or calculating the payoff, and in a repeated game one such action changes the next state and can affect every later round. A plausible final trajectory is therefore not, by itself, evidence that the model understood the game.

Prompt-based evaluation is itself sensitive to presentation choices that do not change what the task asks. Meaning-preserving formatting, option order, response-token identity, output-format requests, whitespace and persona all shift model output [22, 25, 26, 28, 32]. The same threat appears in strategic choice specifically: reversing action labels or reassigning their payoffs changes choice frequencies [17], decisions move with topic, actor and world framing across vignettes built on one underlying game [24], and performance falls when a familiar default mapping is replaced by an explicitly stated alternative [29]. This is why we audit wording, answer order, arithmetic disclosure, narrative framing and opaque response codes separately, and why we do not treat one familiar presentation as a neutral measuring instrument.

Reproducible game-based frameworks approach the problem from the other side. FAIRGAME standardises repeated-game experiments across models, languages and persona conditions and compares them with game-theoretic predictions [8]; later work adds payoff scaling, a Public Goods Game and supervised recognition of canonical strategies from action trajectories, and finds cooperation moving opposite to its evolutionary benchmark [18, 19]. That motivates our use of full trajectories and theoretical benchmarks. Our audit asks the prior question those studies do not: whether the agent can track the race that produced the trajectory at all.

This distinction is easy to lose. An agent can produce an aggregate UNSAFE rate close to the human one while relying on incorrect state computations, prompt-specific heuristics or unstable action encodings. Aggregate similarity is therefore not evidence that an agent understands the task, reproduces human decision dynamics, or holds a general safety preference [7].

We address this problem with an *audit-first evaluation*: we check that the game and the agent’s use of its rules are valid before we interpret the agent’s behaviour. We reproduce the two-player game studied by Domingos and Han [14] and add a compatible multi-player version based on Han et al. [16]. The game engine, not the LLM, resolves every simultaneous round, and we save every prompt, response, parsed action, retry, state change, configuration, random seed and final outcome.

The audit has four levels. First, *mechanical validity* means that the game engine applies the stated rules correctly. Second, *task*

*validity* means that an agent can recall the rules, rebuild the current state, apply a state change, and calculate the final result. Third, *representation robustness* means that choices remain stable when wording or response labels change but the game itself does not. Fourth, we compare valid trajectories with published human patterns. The human data are a reference for observable behaviour. They are not a target that an LLM must copy, and LLM outputs are not treated as replacements for human participants.

Two lessons from LLM-based behavioural simulation shape that boundary. The first is that a plausible output is not a comprehended task: qualitative findings reproduce in three of four experiments while the fourth shows a hyper-accuracy distortion [1], and a model can predict an opponent’s pattern without acting on it [2]. Neither a plausible trajectory nor a verbal strategy report is therefore treated here as evidence of comprehension.

The second is that matching a mean is much weaker than matching a distribution. Simulated populations recover broad contrasts while showing less heterogeneity than the people they stand in for, in markets, cognitive phenotyping, large survey populations and persona-based simulation of perception [9, 12, 13, 15, 27], which is why position papers frame social simulation as defensible for exploratory use until validated against the population of interest [3]. Strategic-game evidence is similarly unsettled, with policies that shift with parameters and framing, heuristics applied more rigidly than people apply them, cooperation in place of the predicted equilibrium, and multi-turn action sequences that resist reproduction from prompts alone [20, 21, 30, 31]. Our human comparison therefore reports within-population diversity, not only aggregate UNSAFE rates.

We ask two questions. Can an agent track the game well enough for its choices to be interpreted strategically at all, and are those choices stable across presentations that leave the game unchanged? And given that answer, how do agents respond to risk, to opponent history and to relative progress, against the game-theory and human benchmarks?

The order matters, because the audit changes what the behaviour means. The tested routes express no common strategic policy: routes with similar aggregate UNSAFE rates respond differently to risk, to opponents and to position, rule recall coexists with incorrect state tracking, and probe accuracy is descriptively associated with how sharply a route responds to risk. Measured directly on action sequences, human participants are more diverse than every route that passes the audit, and the only two routes whose diversity reaches the human range are routes the audit refuses. Our contribution is the combination: an auditable race mechanism, a validity gate applied before any strategic reading, and a comparison against game-theory and human benchmarks made only for the routes that pass it. Exploratory multiplayer results remain in the supplementary material, because the available position analysis covers two checkpoints and cannot identify a general group-size effect.

Each reported behavioural conclusion is scoped to its tested checkpoint, prompt version, decoding contract, experimental configuration, and run period. The labels SAFE and UNSAFE denote actions within the experimental game and should not be interpreted as direct measurements of general LLM safety, subjective risk preference, or suitability for autonomous deployment.

## 2 PRELIMINARIES

We first specify the two-player game reproduced from Domingos and Han [14] (§2.1), then its generalisation to  $N > 2$  companies (§2.2), where only the stage payoff changes shape (Figure 1).

### 2.1 Two-player game

Two companies  $i \in \{1, 2\}$  play a repeated race. In every round  $t$ , both simultaneously choose an action  $a_i^t \in \{S, U\}$  (Safe, Unsafe) and commit to it before either learns the other’s choice, and only then does the round resolve (Figure 1a). Resolution has two parts. First, progress accumulates as  $P_i^t = P_i^{t-1} + \sigma(a_i^t)$  with  $\sigma(S) = 1.0$  and  $\sigma(U) = 1.5$ , so Unsafe always closes the race faster. Second, the round pays a stage payoff  $\pi_i^t = \pi(a_i^t, a_{-i}^t)$ , read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \quad (1)$$

Unsafe strictly dominates Safe at the stage level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon  $\mathbb{E}[T] = 9$ . Neither company is ever told the horizon, so neither can condition on the current round being the last (Figure 1a). Termination resolves in two further steps. First, the company with the higher cumulative progress  $P_i^T$  wins a prize  $B = 100$ , split 50/50 on an exact tie. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \quad (2)$$

where  $n_i^U(T)$  counts company  $i$ ’s Unsafe choices over the whole race and  $p_r^{\max}$  is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against  $q_i(T)$  fails, that company’s entire race payoff, accumulated stage payoffs plus the prize, is zeroed. A company that ends behind is never subject to the draw: it keeps its stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

### 2.2 N-player game

The  $N$ -player game changes exactly one thing. Simultaneous sealed actions, the progress increments, the hidden minimum-five-round horizon, the terminal prize  $B = 100$ , and the risk formula of Equation 2 all carry over unchanged, with the prize split evenly among the companies sharing the highest progress. What cannot carry over is the stage-payoff matrix, because with more than two companies there is no single opponent to index it against. Equation 1 therefore generalises to the group-count rule of Han et al. [16], in which both payoffs depend on how many of the  $N$  companies chose Safe and not on which ones did (Figure 1c). The engine supports  $N \in \{2, 3, 4, 5\}$ . Since the multiplayer runs are an exploratory extension, the rule, its parameters and its results are in the supplementary material.

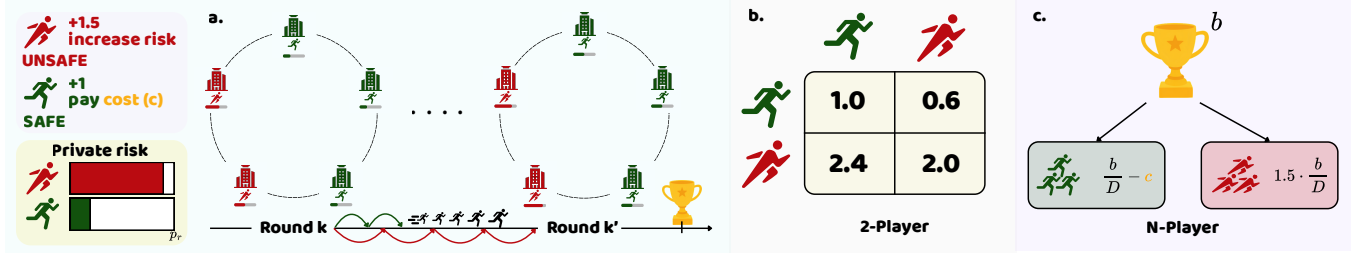


Figure 1: The AI-race mechanism, shared by the two-player and  $N$ -player games. (a) Each round a company advances by choosing Safe (green, slower, pays a cost) or Unsafe (red, faster, adding private setback risk); actions are simultaneous, the stopping time is hidden, and terminal prize  $B$  goes to the progress leader or is split among tied leaders. (b) The two-player stage-payoff matrix, Equation 1. (c) The  $N$ -player stage-payoff rule, stated in full in the supplementary material, in which the per-round benefit  $b$  is distinct from terminal prize  $B$ .

### 2.3 Game-theoretic benchmark

The two incentives oppose each other. Stage-level dominance rewards Unsafe immediately, while across the full race Unsafe choices raise the setback risk for a winner or tied winner, so the best action can depend on the risk condition, earlier actions, and the expected response of other players.

We use four simple strategies as a theory benchmark: Always Safe (AS), Always Unsafe (AU), Conditional Safe (CS), which starts Safe and then copies the opponent, and Conditional Unsafe (CAS), which starts Unsafe and then copies the opponent. In the reconstructed finite-population evolutionary model of Domingos and Han [14], the most common strategy changes from AU at  $p_r^{\max} = 0.10$ , to CAS at 0.60, and to CS at 0.90: the theory predicts not one fixed amount of Unsafe play but a risk-dependent shift between unconditional speed, an aggressive opening and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output, because prompted self-play trajectories are not samples from an evolutionary population.

## 3 EXPERIMENTAL DESIGN AND EVIDENCE POLICY

### 3.1 Agent protocol and units of analysis

At each decision an agent receives a versioned prompt carrying the round, the assigned maximum private risk, the disclosed state, and the preceding revealed action profile when  $t > 1$ . Same-round decisions are sealed: every response comes from the same pre-action snapshot, before the engine resolves actions, progress, payoffs and risk, so the environment rather than generated text determines all transitions. Raw response, parsed action, retry and parser status, pre-turn state, terminal record, configuration and seed allocation are retained for every decision.

Sampling parameters here are provider-visible contracts, not intended settings. The run manifests record a requested temperature of 0.7 that the hosted SDK forwarded on no baseline route, and no provider confirmed an effective value. Seed handling was route-specific: the SDK stripped the requested seed on the Google routes and forwarded it, application unconfirmed, on the OpenAI and Anthropic ones.

Both seats of a race are the same endpoint, so every gameplay rate below is mirror-match play. Against a scripted always-SAFE rival, one admitted route plays UNSAFE on 24.7% of decisions at risk 0.1, where its mirror-match rate is 98.9% (supplementary material): a mirror-match rate is an equilibrium of one policy meeting itself, not a measurement of how it treats risk.

The main behavioural outcome is whether an agent chooses Unsafe. The race, not the individual decision, is the main independent experimental unit, because decisions from one race depend on earlier states and are not separate random samples; uncertainty is grouped by race or matched repetition wherever possible. A *parse failure* is a response that does not follow the required action format, and because the fallback action changes later states, one marks the whole race as contaminated.

### 3.2 Evidence strata and admission

We separate mechanical validity, task validity, diagnostic pilots and confirmatory evidence (Table 1). A *diagnostic pilot* is a small exploratory test for possible effects or failures and does not estimate a stable effect. A *confirmatory* study tests a question under a plan fixed before any result is inspected, and a run enters that stratum only when the protocol, route identifier, prompt and configuration hashes, race count and exclusion rules were all fixed and recorded beforehand.

Two comprehension batteries appear in this paper and should not be confused. An earlier open-source audit uses a 41-probe bank over the same six domains, varying wording, paraphrase, calculator disclosure and answer order, and scoring format compliance separately from semantic correctness. The confirmatory frontier endpoint admission of §4.1 is a separate, frozen battery. A paired behavioural diagnostic compares the canonical prompt against a deterministic four-row decision card that discloses the focal agent’s immediate payoff, progress and private-risk consequence for each action profile, while revealing neither the opponent’s move nor the stopping horizon.

Three further manipulations are described in full in the supplementary material: a persona framing with a length-matched neutral placebo, eight narrative skins that retell the same game as different stories, and a substitution of the opaque response codes  $P$  and  $Q$  for SAFE and UNSAFE. The last two are read three ways:

**Table 1: Evidence strata. Pilot modules are never pooled with confirmatory inference. Admission and the neutral baseline cover all nine routes, of which five are admitted; the crossed representation cell uses two of them, and the matched group-size grid and the scripted-rival campaign use one.**

| Module                 | Primary units | Observed outputs | Use               |
|------------------------|---------------|------------------|-------------------|
| Endpoint admission     | 180 probes    | 540 answers      | confirmatory gate |
| Neutral baseline       | 270 races     | 5,022 decisions  | confirmatory      |
| Crossed representation | 240 races     | 4,464 decisions  | confirmatory      |
| Matched group size     | 120 races     | 3,696 decisions  | confirmatory      |
| Scripted rival         | 120 races     | 1,116 decisions  | confirmatory      |
| Calculator diagnostic  | 60 races      | 1,116 decisions  | paired diagnostic |
| Fixed-state replay     | 1,536 cells   | 1,536 responses  | prompt diagnostic |
| Context pilot          | 768 races     | 13,680 decisions | audit trail only  |
| OpenAI $N=3-5$ pilot   | 180 races     | 6,120 decisions  | exploratory only  |

a paired first-round comparison at an identical state, a *fixed-state replay* that asks for one action at a previously recorded state without letting the answer change later states, and a *live trajectory* in which each answer does change the next state. A comprehension check gates all of them: if it fails, a behavioural difference shows prompt-conditioned output rather than informed play.

## 4 RESULTS

We report the validity gate first, because every strategic claim below depends on it, and then ask how measured comprehension relates to behaviour at all. Three- to five-player races, full model tables, predictive diagnostics and robustness checks are in the supplementary material.

### 4.1 Validity gate: can agents follow the race?

Nine routes were audited under one frozen probe bank of 20 atomic probes at three requested repetitions, 540 retained outputs in total, with no parse failure. Admission requires 80% overall accuracy, at least 12 of 15 state-reconstruction probes (75%), and 75% on terminal scoring. The closed-form expected-payoff probes are recorded but do not gate admission, because the game asks an agent to choose an action, not to solve for an expected value.

Table 2 reports the result, and comprehension is distributed unevenly. Reading the payoff matrix and stating the rules are solved by everyone; every difference between endpoints appears in the harder subtasks, where accuracy at rebuilding the current state from a short history ranges from 100% down to 20%. Five routes clear the gate and four do not.

Two points follow. No route reaches 75% on closed-form expected payoff and five score zero, so the arithmetic gap is general rather than a property of one weak endpoint. And a correctly formatted action carries no information about whether the agent tracked the state behind it: the parser was satisfied on all 540 outputs while state reconstruction varied across the full range, which is the concrete sense in which a valid action is not a valid measurement. The battery has been administered three times, twice under an identical contract; between those two, individual state-reconstruction scores moved by up to 13.3 points, two calls in fifteen, while no verdict changed. A verdict is therefore an admission decision, not a precise estimate of comprehension. Probe-level outputs, retained failure records and unreachable routes are in the supplementary material.

**Table 2: Semantic accuracy on the frontier endpoint admission, one row per audited route, ordered by overall accuracy. Reading the stage-payoff matrix was answered perfectly by every route, and rule recall by every route except GPT-5.4 nano (75.0%), so those two subtasks are omitted here.**

| Route                 | State recon. | State trans. | Term. score | Exp. pay (diag.) | Overall | Adm. |
|-----------------------|--------------|--------------|-------------|------------------|---------|------|
| Gemini 3 Flash        | 93.3         | 100.0        | 100.0       | 50.0             | 93.3    | yes  |
| Claude Opus 5         | 93.3         | 100.0        | 100.0       | 33.3             | 91.7    | yes  |
| GPT-5.4               | 100.0        | 100.0        | 100.0       | 0.0              | 90.0    | yes  |
| GPT-5.5               | 80.0         | 100.0        | 100.0       | 50.0             | 90.0    | yes  |
| Claude Sonnet 5       | 100.0        | 100.0        | 80.0        | 0.0              | 85.0    | yes  |
| Gemini 3.1 Flash-Lite | 73.3         | 100.0        | 80.0        | 16.7             | 80.0    | no   |
| GPT-5.4 mini          | 66.7         | 66.7         | 86.7        | 0.0              | 75.0    | no   |
| Gemini 3.5 Flash-Lite | 60.0         | 50.0         | 60.0        | 0.0              | 65.0    | no   |
| GPT-5.4 nano          | 20.0         | 50.0         | 66.7        | 0.0              | 51.7    | no   |

### 4.2 Is measured comprehension associated with the risk response?

The audit is only worth running if it tells us something about the behaviour it gates. All nine audited routes also completed the confirmatory two-player baseline under one frozen protocol: 30 races, 558 decisions and zero parse failures each, with 10 races in every risk cell. We summarise each route’s strategic sensitivity by its *risk response*, the UNSAFE-rate difference between the lowest and highest assigned private risk, resampled over races because the two seats of a race share its sampled horizon and setback draw.

Two routes also ran in the earlier confirmatory block, a temporal replicate of one frozen protocol, and agree closely: the largest per-cell difference is 1.1 points for Gemini 3 Flash and 5.3 for Claude Sonnet 5, against risk responses of 38.7 and 57.0. Provider-side variation is thus small relative to the reported effect, which matters because no provider confirmed applying the requested seed.

The risk response spans almost the whole available range, from 7.0 percentage points for GPT-5.4 mini to 100.0 for Claude Opus 5 (Figure 2a), and state-reconstruction accuracy is descriptively associated with it, as in a commons game with many more probes [23]. Across the nine routes, the rank correlation between risk response and state-reconstruction accuracy is  $\rho = 0.87$  (exact permutation  $p = 0.004$ ,  $n = 9$ ), rising to  $\rho = 0.92$  ( $p = 0.003$ ,  $n = 8$ ) when Claude Opus 5, whose policy is a step rather than a slope (Figure 2c), is set aside. We report the correlation with and without it, because a single extreme route should not carry a rank statistic computed over nine.

Three cautions belong with that number, and none is removable by more careful statistics. Nine commercial endpoints are not a sample from a population of models, so this is a descriptive association, not an estimated effect. The endpoints differ in far more than their probe scores, so nothing here separates comprehension from provider, scale or training. That confound is legible in the route strings alone: calling a route small when mini, nano or lite appears as a whole word reproduces the gate’s split exactly, 9 of 9, exact permutation  $p = 1/126 = 0.008$  (figure in the supplementary material), so on these nine routes the association cannot be separated from scale or the training differences that travel with it. And the gate does not partition the sample cleanly: Gemini 3.1

**Table 3: Disclosed-arithmetic diagnostic: canonical vs. decision-card conditions, 30 races each. Intervals are 95% percentile bootstrap intervals over the 30 races.**

| Condition     | UNSAFE rate | 95% CI        | Mean payoff |
|---------------|-------------|---------------|-------------|
| Canonical     | 52.0%       | 48.9 to 55.4% | 42.77       |
| Decision card | 60.8%       | 55.1 to 65.3% | 42.21       |

Flash-Lite misses admission on state reconstruction, 73.3% against the 75% threshold, yet responds to risk as strongly as three admitted routes (Figure 2b), so the threshold is a declared entry condition, not a behavioural predictor.

What the pattern does establish is the practical point. The two routes whose play is flattest in risk, GPT-5.4 nano at 9.7 percentage points and GPT-5.4 mini at 7.0, score 20.0% and 66.7% on state reconstruction, both below the gate. Their UNSAFE rates near 60% would sit unremarkably in a cross-model table and are close to the human mean, yet come from endpoints that largely cannot rebuild the state the risk applies to, and no aggregate rate reported without a comprehension gate can tell that apart from a deliberate policy of moderate risk-taking. Across the four refused routes the risk response averages 19.4 points against 54.8 for the five admitted ones.

### 4.3 Strategic robustness under equivalent presentations

We next ask whether sampled behaviour is stable under manipulations that leave the underlying game mechanics unchanged.

*Disclosed arithmetic.* The canonical and decision-card conditions each completed 30 races and 558 decisions with zero parse failures, and matched risk-by-repetition cells had identical realised horizons. Table 3 compares them, with intervals resampled over races rather than decisions because two decisions in one race share a state history. Only 3.3% of paired first-round decisions changed, so the divergence appeared later, after feedback accumulated. Disclosing the current round’s arithmetic shifts UNSAFE play upward by about nine points, but the two race-clustered intervals still touch, so we read it as suggestive rather than established.

*Opaque symbolic mapping.* The earlier local context-skin pilot confounded mapping with repetition parity and failed its comprehension gate, so it is excluded here and kept only in the audit trail. A fully crossed rerun, counterbalancing two narrative skins against the two assignments of the opaque response codes  $P$  and  $Q$  within repetition blocks, completed on both admitted routes: 120 races and 2,232 decisions each, zero parse failures. Each contrast is taken within a repetition block, and the recorded draw hashes confirm that paired cells share one sampled horizon, which is thereby differenced out.

Both presentations move play, the code assignment by a similar amount on the two routes: swapping which code denotes SAFE raises UNSAFE play by 9.5 percentage points on Gemini 3 Flash (95% interval 5.6 to 13.8) and by 8.3 on Claude Sonnet 5 (5.4 to 12.2), over 60 paired blocks each. The narrative skin matters too, but route-specifically: retelling the same game as an abstract contest

rather than a technology race raises UNSAFE play by 10.8 points on Gemini 3 Flash (6.0 to 15.6) and by 3.4 on Claude Sonnet 5 (0.9 to 7.4). No element of the mechanism differs across any of these cells, and the scope is two audited routes, not endpoints in general.

### 4.4 Game theory versus prompted race behaviour

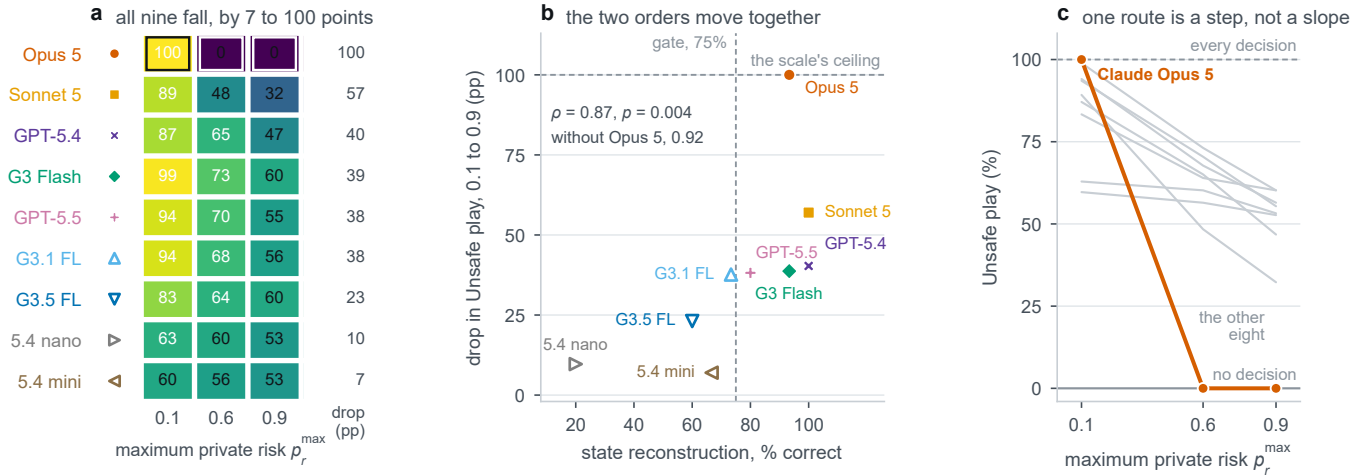
At the reference setting reported by Domingos and Han [14], selection strength  $\beta = 2$  with mutation  $\mu = \beta/Z$ , the reconstructed evolutionary model predicts a switch rather than a gradient: UNSAFE play of 99.2%, 98.0% and 1.9% at maximum risks 0.1, 0.6 and 0.9. The audited routes instead decline smoothly: in the earlier confirmatory block Gemini 3 Flash plays UNSAFE 98.9%, 74.2% and 59.1% and Claude Sonnet 5 89.2%, 46.2% and 37.6%, each over 30 races and 558 decisions (Figure 3).

Read against that single reference setting, the prompted routes look like a failure to reproduce the theory. That reading does not survive a sensitivity check. We swept selection strength across  $\beta \in [0.001, 10]$  crossed with three mutation rules, 90 stationary-distribution cells, each estimated from independent chains with the between-chain spread retained so a badly-mixed cell cannot be mistaken for a result. The sharp switch is a property of strong selection, not of the game: as  $\beta$  falls, the predicted high-risk UNSAFE rate rises off the floor and the risk response becomes graded. At  $\beta = 0.01$  with  $\mu = 0.05$  the same model predicts 87.3%, 63.9% and 38.0%, the closest cell in the entire sweep to both audited routes, within 10.3 percentage points of Claude Sonnet 5 and 15.1 of Gemini 3 Flash in root-mean-square error.

Two things follow. First,  $\beta = 0.01$  and  $\mu = 0.05$  are not parameters we chose to fit: they are the values the source study reports as its own best fit to human play, so the audited routes sit closer to the regime that describes the human participants than to the strong-selection regime usually quoted as the model’s prediction. Second, the interpretation stays bounded. A stationary distribution describes strategy frequencies in a population under selection and mutation, while our measurements are prompted self-play decisions; the agreement in shape says the graded risk response is consistent with a near-neutral, high-exploration population, not that the routes are evolving. What the sweep establishes is narrower and still useful: a claim that LLM agents contradict the evolutionary benchmark cannot rest on one setting of the selection strength, because the sign of the high-risk gap depends on it.

### 4.5 Two-player strategic diversity: humans and LLMs

We compare model trajectories with the human benchmark of Domingos and Han [14] directly on the literal first-five-round action-and-position histories, without first reducing them to hand-designed summary statistics. The comparison pools 420 trajectories from the seven tested models’ two-player baseline, 60 per model, with 340 complete human trajectories from the public de-identified dataset (one incomplete human record excluded),  $N = 760$ . Six of the seven checkpoints carry an endpoint-admission verdict from Table 2; GPT-5-nano is no longer offered on the audited identity and so cannot be gated at all, and is retained here as historical description. Each player is represented by 15 raw features: own



**Figure 2: Measured comprehension against strategic sensitivity over the nine audited routes.** (a) UNSAFE rate per route-by-risk cell, 186 decisions each, with the drop from risk 0.1 to 0.9 at the right; a ringed tile sits on a boundary and a filled glyph marks an admitted route. (b) State-reconstruction accuracy against that drop:  $\rho = 0.87$  over nine routes, 0.92 with Claude Opus 5 set aside. It is an agreement between two measurements of the same nine endpoints, not an estimated effect, since these routes differ in far more than one probe domain. (c) Claude Opus 5 against the other eight in grey: UNSAFE on all 186 decisions at risk 0.1 and SAFE on all 372 above it, a step rather than a slope.



**Figure 3: The evolutionary benchmark and the audited routes differ in shape, not level.** Panel a solves the small-mutation limit on a fine grid of risk values at both declared selection strengths: at  $\beta = 2$  the predicted UNSAFE rate falls 98 points inside one 0.005 step, while eight routes decline gradually. Claude Opus 5, in its own colour, does have a cliff, but no risk level was run inside the interval where it falls. Panel b inverts the model, asking which risk would make it emit each observed rate: the configured range of 0.80 returns as a band of 0.12, so panel c's three configured levels are not separable in its answers. Every curve is the small-mutation limit, which cannot represent the finite mutation rate of the points it is drawn at, so this bounds the regime in which the benchmark is read and is not a test of the model.

action, opponent action, and own-minus-opponent progress gap

entering each of rounds 1 to 5, which are the recorded states and

actions themselves rather than derived quantities. The archetype projection uses 341 participants rather than 340, because its five descriptor summaries can be computed for one further participant whose complete five-round sequence is unavailable.

Mean UNSAFE play over rounds 1 to 5 is lowest for GPT-5-nano (17%), then Claude Sonnet 5 (22%), Claude Opus 5 (33%), GPT-5.4-nano (54%) and humans (56%), with the Gemini models from 73% (Gemini-3-Flash) to 83% (Gemini-3.1-Flash-Lite). No single aggregate rate characterises the tested models, and closeness to the human mean is, on its own, uninformative about anything beyond that one number.

A complementary view in the supplementary material projects each trajectory onto the nearest of four behavioural archetypes fitted to the human sample alone. The human sample occupies every archetype at every cluster count from two to six, while each checkpoint except GPT-5.4 nano is more concentrated, so that conclusion does not depend on the choice of four; the archetypes themselves are exploratory descriptions rather than recovered types, best separated at four groups but only moderately stable under bootstrap. A t-SNE embedding of the same pooled space is there too, and we rest no conclusion on it: recolouring the identical coordinates by outcome rather than by identity separates it almost entirely by UNSAFE rate, so the apparent separation between populations is largely a separation between action rates. The diversity comparison below is therefore made directly on the action sequences, with no embedding and no clustering step.

*How much diversity, measured directly.* We compare the exact paired five-round action sequence of each player, a ten-bit string, within each risk condition; because the progress increments are fixed, that string is the whole trajectory. This comparison reads all nine audited routes on the confirmatory baseline, 60 trajectories per route, not the seven-checkpoint pilot export used above. Figure 4 counts how many distinct sequences each population uses out of twenty, against the distribution of 20,000 draws of twenty human participants matched on sample size and risk condition; the human sample is five to seven times larger, so without matching a population could look more diverse merely for being bigger. The supplement adds two rarefied statistics, since either alone can mislead: Hill  $q=1$  counts how many sequences a population effectively produces, the mean pairwise Hamming distance how far apart they are.

The result is not a low percentile but the absence of one. Twenty-one of the twenty-seven route cells fall below every one of the 20,000 human draws; against a stricter null of ten complete dyads, which matches the routes’ unit of independence and not only their sample size, eighteen do, and never fewer than sixteen over forty redraws of that null. Humans reach an effective 18.9, 19.7 and 19.3 sequences out of 20 at risk 0.1, 0.6 and 0.9, at mean pairwise distances of 0.47, 0.50 and 0.50. The compression is severe rather than marginal: Claude Opus 5 produces a single sequence, repeated by all 20 of its trajectories, at every risk level, and Claude Sonnet 5 uses two at risk 0.1. A population of twenty agents collapsing to one policy is not a narrower distribution of strategies; it is the absence of one.

Two checkpoints are the exception, and they are the most informative rows in the figure. GPT-5.4 mini uses 20, 17 and 17 distinct

sequences out of 20 at risk 0.1, 0.6 and 0.9, and GPT-5.4 nano 16, 19 and 16; both land inside the human null at every risk level, and no other route reaches it once. On this measure they are not less diverse than people. Neither passes the gate of Table 2: GPT-5.4 nano scores 51.7%, the lowest of the nine, and GPT-5.4 mini 75.0%. Every checkpoint that passes the gate falls below every human draw at every risk level, and the rarefied Hill  $q=1$  interval puts all five entirely below the human one too. The exception is two routes rather than one and the gate refuses both, which strengthens the argument. The audit is therefore not a formality placed in front of the interesting result; it is what separates a population whose variety reflects strategy from one whose variety is consistent with not tracking the game. Reported without the gate, these two rows would read as the headline finding that small models match human strategic diversity.

A shallow classifier trained on the same raw sequences, with folds grouped by race, recovers the producing population well above a grouped permutation null (balanced accuracy  $42.3\% \pm 4.3$  against a null mean of 12.4%,  $p < 0.001$ ), but the groups overlap: that is evidence of observable trajectory differences, not of a recoverable latent strategy.

This heterogeneity is not merely a visual impression. Fitting three nested logistic models to the pooled neutral-baseline pilot (risk-only, LLM-identity-only, and LLM-identity-by-risk interaction) shows that allowing the risk-response profile to vary by LLM improves fit substantially beyond LLM-specific intercepts alone,  $\chi^2(10) = 354.7$ ,  $p < 10^{-3}$ , with several risk-by-LLM cells approaching complete separation. A convergence warning in the interaction model means that the displayed likelihood-ratio statistic is approximate, so we report the conservative threshold rather than false precision. The fit covers the five-model neutral baseline only; the two Claude models are not in it. Neither caveat changes the conclusion: no single risk-response curve represents all five, so cross-model differences are differences in *shape*, not only in level.

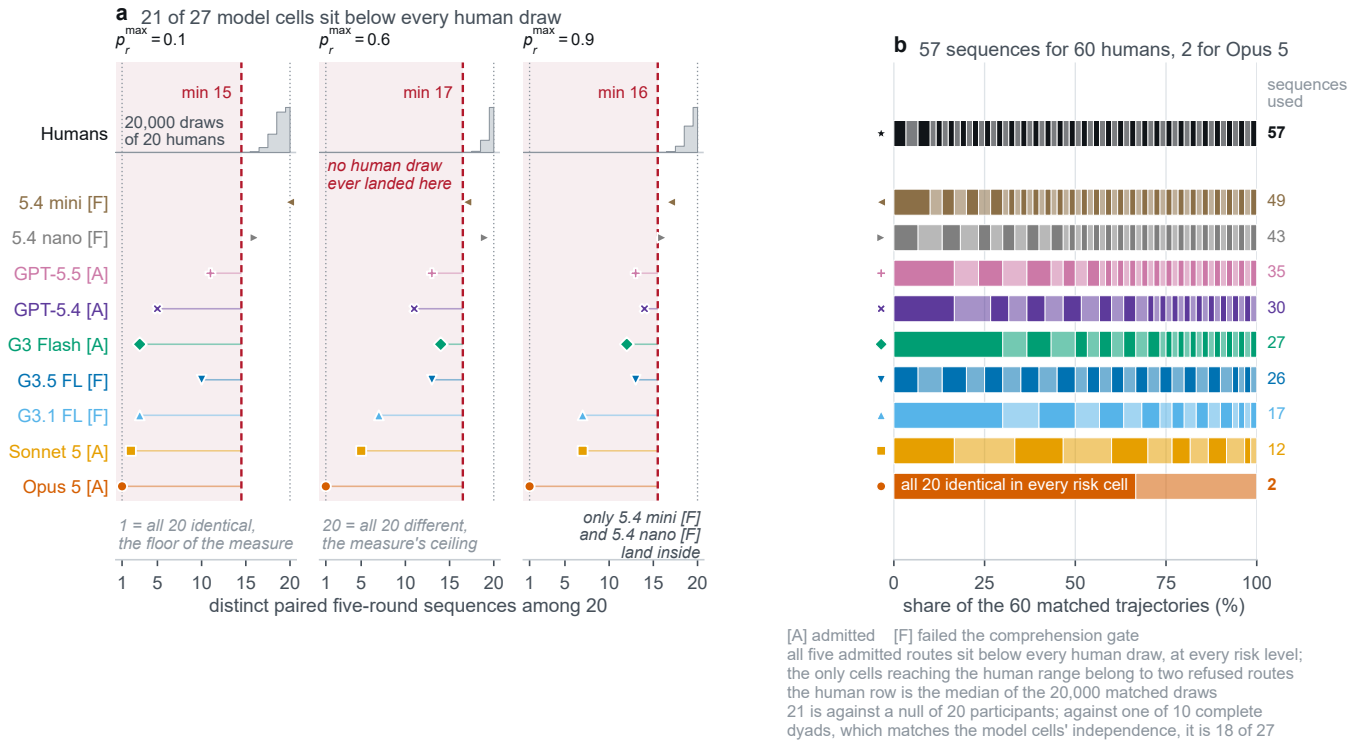
Raw player-level UNSAFE rates, plotted per risk condition in the supplementary material, show the same picture: human rates spread across nearly the whole available range at every risk level while each checkpoint occupies a narrow band of its own. Similarity between a checkpoint’s aggregate UNSAFE rate and the human mean therefore does not imply that it reproduces the human distribution.

Which pre-decision variables predict an UNSAFE choice is asked separately in the supplementary material, as association and not cause: the human data are organised mainly by the opponent’s previous action, 56% of total SHAP magnitude, while the checkpoints share no such pattern, so two populations can have similar UNSAFE rates while following different response rules.

## 5 DISCUSSION AND CONCLUSION

The lesson is not that an audited route “understands the game”. Admitted routes recall the rules, rebuild the tested states and apply terminal eligibility, yet none reliably computes a closed-form expected payoff, and the routes that fail the gate still return a perfectly formatted action every time. A study that reports action rates without per-route comprehension accuracy and parser health cannot tell those situations apart, and they license opposite conclusions.





**Figure 4: Human participants use far more of the policy space than every checkpoint the gate admits. All nine audited routes are drawn, on the confirmatory baseline. Panel a places each checkpoint’s count of distinct paired five-round action sequences against 20,000 draws of twenty human participants, matched on sample size and risk condition; the shaded region is the range no human draw reached. Panel b shows the mechanism, which sequences each population actually uses. Marks give each checkpoint’s verdict under the admission gate of Table 2, and the two checkpoints that reach the human range are both checkpoints it refused.**

One contrast is worth keeping in view, because it is so often conflated. The human participants’ incentivised pre-game risk score has almost no relation to their later UNSAFE play ( $r = -0.015$ ,  $p = 0.79$ ,  $n = 341$ ) and does not even separate the behavioural archetypes, whereas sweeping a prompted risk persona from its lowest to its highest level moves UNSAFE play by 50.5 to 98.3 percentage points. A measured preference describes a person before play; a persona label rewrites the decision context during it, so a persona is a strong policy prompt and not evidence of a stable risk preference. The curves are in the supplementary material.

The human experiment is an external behavioural reference, not a target an agent should reproduce. Where a route matches a human association, the two populations behave similarly on that one measured aspect of this task; it does not follow that they share a mechanism, and a mismatch does not invalidate the human result. Every comparison here is scoped to its tested route, prompt version, decoding contract, provider path and run period.

Three limitations bound the claims directly. The comprehension-to-behaviour association covers nine endpoints differing in far more than their probe scores, so it cannot separate comprehension from provider, scale or training. The representation-robustness design is crossed for two routes, enough to show the effect replicates but not to characterise endpoints in general. And the evolutionary

comparison is a reconstruction audited against pinned EGTtools source rather than a bitwise reproduction of undisclosed author code, so it bounds the regime in which the benchmark is read rather than fitting the agents.

Four further limits are worth stating. Fixed-state replay and live play measure different things, one decision at a saved state against dependent decisions along a trajectory, so their effect sizes are not interchangeable. A matched group-size grid now exists on one route, but adding a company changes the stage payoff and the prompt length too, so no pure group-size effect is identifiable. Seed forwarding is provider-specific: one route’s SDK stripped the requested seed and another forwarded it unconfirmed. And this game models a private speed-versus-safety trade-off and nothing else, with no collective harm, regulation, communication or uncertainty about real capabilities.

Strategic safety behaviour in these races is not a fixed property of an LLM: across nine audited routes, rule recall does not ensure state tracking and an equivalent presentation can change later play. The audit is what makes those differences readable, since the routes that fail it include both the one closest to human diversity and the ones nearly flat in risk.



## REFERENCES

- [1] Gati V Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, Honolulu, Hawaii, USA, 337–371. <https://proceedings.mlr.press/v202/aher23a.html>
- [2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (01 Jul 2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [3] Jacy Reese Anthis, Ryan Liu, Sean M. Richardson, Austin C. Kozlowski, Bernard Koch, Erik Brynjolfsson, James Evans, and Michael S. Bernstein. 2025. Position: LLM Social Simulations Are a Promising Research Method. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*, PMLR, Vancouver, Canada, 81005–81034. <https://proceedings.mlr.press/v267/anthis25a.html>
- [4] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (2023), 337–351. doi:10.1017/pan.2023.2
- [5] Stuart Armstrong, Nick Bostrom, and Carl Shulman. 2016. Racing to the precipice: a model of artificial intelligence development. *AI & SOCIETY* 31, 2 (01 May 2016), 201–206. doi:10.1007/s00146-015-0590-y
- [6] Amanda Askell, Miles Brundage, and Gillian K. Hadfield. 2019. The Role of Cooperation in Responsible AI Development. arXiv:1907.04534 [cs.CY] <https://arxiv.org/abs/1907.04534>
- [7] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis* 32, 4 (2024), 401–416. doi:10.1017/pan.2024.5
- [8] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. 2025. FAIRGAME: a Framework for AI Agents Bias Recognition using Game Theory. arXiv:2504.14325 [cs.AI] doi:10.48550/arXiv.2504.14325 Preprint, version 5 revised March 2026.
- [9] Yang Cao, Haijiang Liu, Arnav Arora, Isabelle Augenstein, Paul Röttger, and Daniel Hershcovich. 2025. Specializing Large Language Models to Simulate Survey Response Distributions for Global Populations. In *Proceedings of NAACL 2025*. Association for Computational Linguistics, Albuquerque, New Mexico, 3141–3154. doi:10.18653/v1/2025.naacl-long.162
- [10] Stephen Cave and Seán S. ÓhÉigeartaigh. 2018. An AI Race for Strategic Advantage: Rhetoric and Risks. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (New Orleans, LA, USA) (AIES '18). Association for Computing Machinery, New York, NY, USA, 36–40. doi:10.1145/3278721.3278780
- [11] Theodor Cîmpeanu, Francisco C. Santos, Luís Moniz Pereira, Tom Lenaerts, and The Anh Han. 2022. Artificial intelligence development races in heterogeneous settings. *Scientific Reports* 12, 1 (02 Feb 2022), 1723. doi:10.1038/s41598-022-05729-3
- [12] Julian Coda-Forno, Marcel Binz, Jane X. Wang, and Eric Schulz. 2024. CogBench: A Large Language Model Walks into a Psychology Lab. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 235)*, PMLR, Vienna, Austria, 9076–9108. <https://proceedings.mlr.press/v235/coda-forno24a.html>
- [13] R. Maria del Río-Chanona, Marco Pangallo, and Cars Hommes. 2025. Can Generative AI Agents Behave Like Humans? Evidence from Laboratory Market Experiments. arXiv:2505.07457 [econ.GN] <https://arxiv.org/abs/2505.07457> Preprint.
- [14] Elías Fernández Domingos and The Anh Han. 2026. Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment. arXiv:2607.26034 [cs.AI] <https://arxiv.org/abs/2607.26034>
- [15] Salvatore Giorgi, Tingting Liu, Ankit Aich, Kelsey Jane Isman, Garrick Sherman, Zachary Fried, João Sedoc, Lyle Ungar, and Brenda Curtis. 2024. Modeling Human Subjectivity in LLMs Using Explicit and Implicit Human Factors in Personas. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 7174–7188. doi:10.18653/v1/2024.findings-emnlp.420
- [16] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. 2020. To regulate or not: A social dynamics analysis of an idealised ai race. *Journal of Artificial Intelligence Research* 69 (2020), 881–921. doi:10.1613/JAIR.1.12225
- [17] Nathan Herr, Fernando Acero, Roberta Raileanu, María Pérez-Ortiz, and Zhibin Li. 2024. Are Large Language Models Strategic Decision Makers? A Study of Performance and Bias in Two-Player Non-Zero-Sum Games. arXiv:2407.04467 [cs.AI] <https://arxiv.org/abs/2407.04467>
- [18] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoai Tran-Le, Alessio Buscemi, Le Hong Trang, and The Anh Han. 2025. Understanding LLM Agent Behaviours via Game Theory: Strategy Recognition, Biases and Multi-Agent Dynamics. arXiv:2512.07462 [cs.MA] doi:10.48550/arXiv.2512.07462 Preprint.
- [19] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoai Tran-Le, Alessio Buscemi, Le Hong Trang, and The Anh Han. 2026. Payoff Scaling Shapes Cooperation in LLM Agents across Languages. arXiv:2601.19082 [cs.AI] doi:10.48550/arXiv.2601.19082 Preprint, version 2 revised June 2026.
- [20] Yuxuan Lu, Jing Huang, Yan Han, Bingsheng Yao, Sisong Bei, Yaochen Xie, Yisi Sang, Qi He, and Dakuo Wang. 2026. Can LLM Agents Simulate Multi-Turn Human Behavior? Evidence from Real Online Customer Behavior Data. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, San Diego, California, USA, 43961–43977. doi:10.18653/v1/2026.acl-long.2034
- [21] Saptarshi Pal, Abhishek Mallela, Christian Hilbe, Lenz Pracher, Chiyu Wei, Feng Fu, Santiago Schnell, and Martin A. Nowak. 2026. Strategies of Cooperation and Defection in Five Large Language Models. arXiv:2601.09849 <https://arxiv.org/abs/2601.09849>
- [22] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, Mexico City, Mexico, 2006–2017. doi:10.18653/v1/2024.findings-naacl.130
- [23] Giorgio Piatti, Zhijiang Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., Red Hook, NY, USA. doi:10.52202/079017-3548
- [24] Isaac Robinson and John Burden. 2025. Framing the Game: How Context Shapes LLM Decision-Making. arXiv:2503.04840 [cs.CL] <https://arxiv.org/abs/2503.04840>
- [25] Abel Salinas and Fred Morstatter. 2024. The Butterfly Effect of Altering Prompts: How Small Changes and Jailbreaks Affect Large Language Model Performance. arXiv:2401.03729 [cs.CL] <https://arxiv.org/abs/2401.03729>
- [26] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying About Prompt Formatting. In *International Conference on Learning Representations*. OpenReview.net, Vienna, Austria. [https://proceedings.iclr.cc/paper\\_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html)
- [27] Jia Wang, Ziyu Zhao, Tingjuntao Ni, and Zhongyu Wei. 2025. SocioBench: Modeling Human Behavior in Sociological Surveys with Large Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Suzhou, China, 26257–26289. doi:10.18653/v1/2025.emnlp-main.1335
- [28] Sheng-Lun Wei, Cheng-Kuang Wu, Hen-Hsen Huang, and Hsin-Hsi Chen. 2024. Unveiling Selection Biases: Exploring Order and Token Sensitivity in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 5598–5621. doi:10.18653/v1/2024.findings-acl.333
- [29] Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Naojung Kim, Jacob Andreas, and Yoon Kim. 2024. Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Mexico City, Mexico, 1819–1862. doi:10.18653/v1/2024.naacl-long.102
- [30] Jiayi Yao, Cong Chen, and Baosen Zhang. 2026. Competition and Cooperation of LLM Agents in Games. arXiv:2604.00487 <https://arxiv.org/abs/2604.00487>
- [31] Kehan Zheng, Jinfeng Zhou, and Hongning Wang. 2025. Beyond Nash Equilibrium: Bounded Rationality of LLMs and Humans in Strategic Decision-Making. arXiv:2506.09390 <https://arxiv.org/abs/2506.09390>
- [32] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. 2024. When “A Helpful Assistant” Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 15126–15154. doi:10.18653/v1/2024.findings-emnlp.888